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Manual Laterality for Simple Reaching and Bimanual Coordinated Task in Naturalistic Housed *Pan troglodytes*

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Abstract Researchers have shown renewed interest in the study of manual lateralization in chimpanzees. Currently there is no consensus confirming the presence or absence of manual dominance at a species level, mainly for populations in the wild and in semicaptivity. We aimed to evaluate the manual laterality in a group of chimpanzees in an intermediate setting (semicaptivity) via 2 tasks: one simple and unimanual (simple reaching) and the other complex and bimanual (tube task). We replicated the same experiments from Hopkins in a new and different sample of chimpanzees. In simple reaching, the hand is used to gather food and the type of grip and the posture are evaluated. The tube task assesses the hand used to extract food from the tube and the method of extraction (digital or instrumental). Through the handedness index we observed that the subjects show clear and strong individual preferences for both tasks (100% lateralized subjects in the tube task; 86% in simple reaching), although we did not detect population preferences for any of the tasks. However, considering both tasks jointly (multiple evaluation), it was possible to detect, for the first time, skilled manual dominance at a group level in semicaptive

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chimpanzees in one multitask index and borderline significance in a second multitask index.

Keywords complex task · hand preference · naturalistic housed · *Pan troglodytes* · replication · simple task

Introduction

From a behavioral aspect, handedness is the functional asymmetry more clearly established in humans. Approximately 85–90% of humans show right-handedness (Annett 2002). Manual lateralization appears to be an indicator of brain hemispheric specialization for motor tasks, and indirectly it would provide information about language functions of the human brain. Some authors have argued that the relation between handedness and brain specialization for language functions suggests unique coevolution in the human brain (Corballis 2002). Accordingly, the evolution of manual dominance in humans would be related not only to development of oral communication but also to nonverbal communication, the use and manufacture of tools, and other cognitive functions (Bradshaw and Rogers 1993). Therefore, manual laterality research in primates is extremely important in tackling comparative studies concerning its evolution in humans.

Results in this area usually indicate 2 general levels of hand preferences: individual and population or group level. According to the former, hand preference emerges when an individual uses one hand more than the other for a significant number of events. At the second level, handedness emerges when most of the sample uses, preferentially, one hand for a single or a variety of tasks. However, there is still considerable controversy regarding hand preference at the population level in nonhuman primates (McGrew and Marchant 1997; Papademetriou *et al.* 2005). Results provided during the last few decades regarding hand laterality in chimpanzees (*Pan*) and the other great apes (*Gorilla* and *Pongo*) do not clearly support the presence of right handedness in these genera. At present, there are clearly 2 conflicting arguments. One of them supports right handedness for chimpanzees (Hopkins 2006), and the other rejects manual asymmetry for these species (Annett 2006; McGrew and Marchant 1997; Palmer 2002). As the leading proponents of these views suggest, i.e., Hopkins on the one hand and McGrew on the other, one can offer numerous explanations for the differences (Hopkins 2006; McGrew and Marchant 1997). They refer to factors such as the environment, methodology, statistical analysis, or the task.

It is now widely accepted that chimpanzees can show manual asymmetry at the individual level in a great variety of simple (Colell *et al.* 1995a; Hopkins and Pearson 2000) and complex (McGrew *et al.* 1999; Sugiyama *et al.* 1993) tasks in different environments: captivity, the wild, and intermediate conditions (semi-captivity or semifree). The main problem is accepting that the asymmetries are applicable at the population level. With respect to the study of laterality in simple motor actions, researchers have demonstrated population preferences only in captivity, finding a greater preference for right- (Colell *et al.* 1995a; Hopkins and

Bard 2000; Hopkins *et al.* 2002; Hopkins and Fernández-Carriba 2000; Hopkins *et al.* 2005b) over left-handedness (Hopkins *et al.* 1993; Morcillo *et al.* 2006). To date, researchers have observed no population preference in the wild (Boesch 1991; Marchant and McGrew 1996; McGrew and Marchant 2001; Sugiyama *et al.* 1993) or in intermediate settings (Fletcher and Weghorst 2005; Mosquera *et al.* 2007) in these tasks. With regard to complex motor activities, the majority of evidence in captivity indicates right-handedness (Colell *et al.* 1995b; Hopkins and Cantero 2003; Hopkins and Pearson 2000; Hopkins and Rabinowitz 1997; Hopkins *et al.* 2007; Hopkins *et al.*, 2004; Wesley *et al.* 2002). In wild populations the only evidence available refers to instrumental tasks, i.e., using an outward element of the body to perform an action (Boesch 1991; Lonsdorf and Hopkins 2005; Matsuzawa *et al.* 2001; McGrew *et al.* 1999) and bimanual grooming (Hopkins *et al.* 2007), whereas in intermediate (Fletcher and Weghorst 2005; Mosquera *et al.* 2007), to date the evidence is inexistent. For example, in a recent paper, Lonsdorf and Hopkins (2005) found not only a left-hand advantage in termite fishing but also a right-hand advantage for leaf sponging and a trend toward a right-hand advantage in wadge dipping. Despite the apparent disparity in the results, particularly with regard to the differences between captivity and the wild, we believe that it should be possible to perform a series of actions to resolve the discrepancy: 1) investigating intermediate situations (Fletcher and Weghorst 2005; Mosquera *et al.* 2007); 2) testing different experimental tasks in the same sample: a task of simple reaching (unimanual task) and a coordinated bimanual task (tube task) (Toback and Hopkins 2006; Wesley *et al.* 2002); and 3) replicating the same experiments in different colonies of animals. We aimed to replicate and extend findings from the simple reaching and tube task experiments of Hopkins in a new colony that is different from the Yerkes studies, instead of resolving the disparity concerning the rearing factor. This is important because it could rule out the notion that results are specific to the Yerkes chimpanzees or due to the observer bias in the Hopkins group. For example, Palmer (2002) and Crow (2005) have both suggested that all the evidence of handedness for the tube task have come from a single laboratory (Hopkins'). In this sense, we assess hand preferences in a group of chimpanzees in a naturalistic environment (intermediate) via 2 experimental tasks: one simple and unimanual (simple reaching) and the other complex and bimanual (tube task). Our principal working objectives have been to obtain the degree and direction of laterality and assess consistency between tasks; to ascertain if lateralization occurs at the group level; and to test other variables associated with the sample and the task.

Methods

Subjects and Housing

We studied 3 captive groups of chimpanzees (*Pan troglodytes*) 4–51 yr at Mona Foundation Primate Rescue Center (Girona, Spain). Animals were in their social groups during the test sessions. The first group (A) comprised 4 males; the second group (B) comprised 4 males and 3 females; and third group (C) comprised 3

males. Table I contains individual information, characteristics, and background of subjects before their arrival at Mona. The subjects were housed in a naturalistic outdoor 5640 m^2 enclosure with wood structures and natural ground with Mediterranean and riverside vegetation. Further, the center has 2 socialization enclosures each of 25 m^2, connected to a 140 m^2 primate pavilion.

Procedure

Two tests evaluated hand preference. The first was based on the tube task proposed by Hopkins (1995), as a bimanual coordinated task sensitive to determine hand motor bias. The second is an experimental simple reaching task that involves simple motor actions.

Llorente performed all the observations and conducted all the experiments (January 2007–April 2007).

Experiment 1: Tube Task

For the tube task, we used cylindrical rubber hoses 25–25.5 cm long and 2.5 cm in diameter. Hopkins showed the usefulness of the tube task as a manual laterality indicator in several studies with captive chimpanzees (Hopkins *et al.* 2003a; Hopkins *et al.* 2004), and researchers have also used it with other primate species such as *Cebus apella* (Lilak and Phillips 2008), *Cebus capucinus* (Meunier and Vauclair 2007) *Cercopithecus neglectus* (Schweitzer *et al.* 2007), *Papio anubis* (Vauclair *et al.* 2005), and other ape species (Hopkins *et al.* 2003b). We filled rubber tubes with food —honey, peanuts, muesli, seeds, etc.— thus preventing its extraction with the tongue or by hitting it. Subjects had to remove the food with their fingers or using tools such as sticks, branches, or canes to facilitate extraction. We used focal animal sampling (Altmann 1974); the session continued until the individual lost interest or left the hose for 1 min. We tested each individual 4 times and separated test sessions by 2 d minimum between sessions. Each individual needed to obtain ≥50 responses. The test was valid if the individual obtained ≥6 responses. We recorded the hand used to extract the food each time the subject inserted its finger/tool, removed the food, and placed its finger/tool in its mouth. We also recorded the finger used to remove the food (thumb, index, middle, ring, little finger).

Experiment 2: Simple Reaching

For the simple reaching task, we used the food supplied daily to the subjects at 13:30 h, as a snack. The snack consisted of muesli, sunflower seeds, peanuts, bread, rice flakes, and apples. Every item was 3×3 cm maximum, enhancing fine precision manipulation. Vauclair *et al.* (2005) criticized the usefulness of simple reaching as the handedness task for its weak enhancement of lateral asymmetries, although other authors have shown it usefulness in testing hand preferences and the effect of grip morphology and posture (Hopkins *et al.* 2005b). We consider it to be a complementary control task (low-level task) to compare the results obtained in more complex bimanual tasks (high-level task).

Table I Individual information, characteristics, and background of chimpanzee sample

Name	Sex	Birth date	Age class	Birth place	Year arrived at Mona Foundation	Background	Previous human contact	Group
Toto	Male	1956	Adult	Wild, Nigeria	2003	Pet, zoo	Half	B
Romie	Female	1979	Adult	Wild	2001	Reproduction, circus	Very high	B
Julio	Male	1979	Adult	Wild?¿	2005	Zoo	Half	C
Víctor	Male	1982	Adult	Wild, Côte d'Ivoire	2006	Pet	Very high	C
Toni	Male	1983	Adult	Wild, Guinea Bissau	2001	Zoo, advertising	Very high	A
Marco	Male	04-06-1984	Adult	Captivity	2001	Advertising, circus	Very high	A
Tico	Male	1985	Adult	Captivity?	2005	Pet, Zoo	Half	C
Charly	Male	02-07-1989	Adult	Captivity	2001	Advertising, circus	Very high	A
Pancho	Male	27-05-1990	Adult	Captivity	2001	Advertising, circus	Very high	A
Waty	Female	1996	Adolescent	Captivity	2002	Circus	Very high	B
Sara	Female	1998	Adolescent	Captivity	2004	Pet, TV	Very high	B
Bongo	Male	2000	Juvenile	Captivity	2002	Circus	Very high	B
Nico	Male	2001	Juvenile	Captivity	2004	Pet	Very high	B
Juanito	Male	2003	Juvenile	Captivity	2003	Pet	Very high	B

A keeper distributed the food to the subjects in their social groups; food was scattered directly on the ground; we provided enough food for all the individuals with the aim of avoiding dominant-subordinates conflicts. The observation session continued until the subject performed ≥100 manual events of simple reaching, as similar studies have done (Vauclair *et al*. 2005). We recorded the hand used to reach the food each time the individual performed a manual grip, followed or not by putting the item into the mouth. We also recorded the grip morphology (thumb-index, middle-index, and others) and posture (quadrupedal, sitting, bipedal, and prostrate) to evaluate its potential effect on hand preference. We recorded an event only when the individual reached the item placed in the central or contralateral side.

Procedural Considerations

Some authors have criticized that recording behaviors as events cause a problem in the data independence (McGrew and Marchant 1997). According to these critics, recording repetitive motor actions of the subjects could increase the sample size and would thus increase the probability of detecting significant asymmetries in the hand use at the individual level. However, other authors failed to detect significant differences in recording manual behaviors as events or bouts, both being sensitive to detecting lateral biases (Damerose and Hopkins 2002; Hopkins *et al.* 2005a; Hopkins *et al.* 2001; Vauclair *et al.* 2005). We chose to record the data as events, i.e., considering the total frequency of manual use. In this way, we can replicate and compare our results with those of most other experimental studies because they follow this procedure.

Data Analysis

We used a methodology similar to that of other authors for similar experiments (Hopkins *et al.* 2004). We performed data analyses at the individual and at the group level. At the individual level, first, we calculated binomial tests on data for each individual to evaluate if they were significantly lateralized. Second, we calculated a handedness index (HI) suggested by Hopkins (1995) for each subject and for each task to quantify individual laterality bias. We calculated the measure via the formula R - L/R+L, wherein R=number of times right hand was used; L=number of times left hand was used. Positive values show right-hand bias and negative values show left-hand bias. For the tube task, we calculated an individual HI adding the total frequencies obtained throughout 4 tests sessions mixed in TUBE (SUM-HI), following the same formula for HI. We used absolute values of HI (ABS-HI) as a reflection of the strength of hand preference, independent of the direction. We calculated a mean HI averaging the HI values (MHI) of 4 tube task tests. To evaluate population level handedness we used a one sample *t*-test based on individuals HI values. We used nonparametric tests (χ^2) to evaluate the proportion of right, left, or nonpreferent individuals. We used the Spearman correlation test to evaluate the consistency of hand preference throughout test sessions in the tube task. Wilcoxon tests assessed differences in direction and strength of hand preferences among tasks. We did not perform group analysis (sex, place of birth, group, or age class) because the sample sizes are too small to make any reasonable comparison and they would reflect random sampling.

Results

Experiment 1: Tube Task

Based on binomial tests, all individuals are lateralized for this task: 10 (71.43%) are classified as right-handed and 4 (28.57%) as left-handed. The average for SUMHI (M-SUMHI) is 0.352 (SE = 0.211); therefore, subjects expressed a right-hand bias. SUMHI and MHI values correlate highly: $R_s = 0.965$; $p = 0.000$; showing that both kinds of handedness characterization were consistent among subjects. Hence we used SUMHI values to minimize type I error. The average for ABSHI for all subjects was 0.811 (SE = 0.058), i.e., the strength of hand preference was high in the sample. The ABSHI values among right- (0.814) and left-handed (0.803) subjects were similar ($U = 15$; $p = 0.478$); hence the strength of hand preference did not vary between groups. There were no differences between the number of right- and left-handed subjects (χ^2 (1, $n = 14$) = 2.571, $p = 0.109$); i.e., there is no population level handedness in our chimpanzee sample. One sample *t*-test did not reveal significant differences either (t (13) = 1.670; $p = 0.119$). However, for test 2 (t (13) = 2.636; $p = 0.021$). and test 3 (t (13) = 2.234; $p = 0.044$) there is a right-hand preference at the group level (Table II), and in test 1 there is a borderline significant right-handedness (t(13) = 1.939; $p = 0.075$).

The subjects removed the food 80.64% with their fingers and 19.36% with tools. They performed most of the actions with the index finger (D2, 63.83%), followed by the thumb (D1, 7.96%), little finger (D5, 6.57%), and middle finger (D3, 2.28%). They performed no action with the ring finger. We found an effect of extraction technique on the hand (D1; D2; D3; D5; TOOL; $\chi^2 = 350.43$; $p = 0.000$). The analyses of adjusted residuals showed that subjects performed extractions with the thumb (AR = 3.9) and index finger (AR = 8.8) preferentially with the right hand and extractions with the middle (AR = 8.2) and little (AR = 16.1) fingers preferentially with the left hand. Tool use seems not to have influenced hand bias (AR = ±2). There was no difference between digital and tool extraction techniques in the direction (HI Finger = 0.333, HI Tool = 0.468; $Z = -0.730$, $p = 0.465$) or strength (ABSHI Finger = 0.827, ABSHI Tool = 0.798; $Z = -0.730$, $p = 0.465$) of hand preference. Four TUBE tests showed a high correlation, indicating that preferences were stable and consistent throughout sessions at the individual level (Table III).

Experiment 2: Simple Reaching

Based on binomial tests, 12 individuals were lateralized and 2 were not: we classified 9 (64.29%) as right-handed, 3 (21.43%) as left-handed, and 2 (14.29%) as nonpreferent subjects. Ten subjects showed HI positive values and four HI negative values. The mean HI for the simple reaching tasks is 0.221 (SE = 0.133). The mean ABSHI for the REACH task is 0.464 (SE = 0.071). There were differences between the number of lateralized and nonlateralized individuals (χ^2 (1, $n = 14$) = 7.143, $p = 0.008$). There was no difference between the number of right-handed and left-handed subjects (χ^2 (1, $n = 14$) = 3.000 $p = 0.083$). One sample *t*-test did not reveal significant differences at the group level (t(13) = 1.654; $p = 0.122$).

Table II Laterality indices for tube task, simple reaching, and multitask evaluation

		Subjects								
Task evaluation	Value	Bongo	Charly	Julio	Juanito	Marco	Nico	Pancho	Romie	Sara
Tube task		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	Individual preference	R	R	R	R	R	R	R	L	L
	SUMHI	1.000	0.922	0.804	0.647	0.858	0.903	0.793	-0.978	-0.931
	M-HI	1.000	0.943	0.768	0.711	0.764	0.965	0.840	-0.917	-0.583
	ABSHI	1.000	0.922	0.804	0.647	0.858	0.903	0.793	0.978	0.931
	% right-hand use	100.00	96.11	90.18	82.35	92.90	95.16	89.66	1.09	3.45
	n responses	165	386	224	28	155	124	319	92	58
	HI test 1	1.000	1.000	0.889	0.392	0.571	1.000	0.882	-1.000	-0.333
	HI test 2	1.000	1.000	0.756	0.500	1.000	1.000	0.795	-1.000	0.000
	HI test 3	1.000	0.945	0.806	1.000	0.905	0.860	1.000	-0.667	-1.000
	HI test 4	1.000	0.828	0.619	0.952	0.579	1.000	0.684	-1.000	-1.000
	HI Finger use	1.000	0.908	0.804	0.665	0.792		0.969	-0.977	-0.931
	HI Tool use		0.975			1.000	0.934	0.524		
	ABSHI Finger use	1.000	0.908	0.804	0.665	0.792		0.969	0.977	0.931
	ABSHI Tool use		0.975			1.000	0.934	0.524		
	Number of techniques used	1	2	2	2	2	1	4	2	2
Simple reaching	Binomial test	0.000	0.000	0.000	0.850	0.000	0.000	0.000	0.000	0.042
	Individual preference	R	L	R	A	R	R	R	L	R
	HI	0.654	-0.481	0.467	0.027	0.367	0.890	0.624	-0.653	0.197
	ABSHI	0.654	0.027	0.890	0.197	0.089	0.480	0.481	0.367	0.624
	% right hand use	82.69	25.96	73.33	51.35	68.33	94.49	81.20	17.33	59.83
	n responses	156	104	150	111	120	127	117	150	117
	HI quadrupedal	0.684	0.027	0.333	0.067	0.256	0.922	0.514	-0.158	0.026
	HI sitting	0.662	-0.731	0.475	-0.071	0.571	0.816	0.814	-0.725	0.512
	ABSHI quadrupedal	0.684	0.027	0.333	0.067	0.256	0.922	0.514	0.158	0.026
	ABSHI sitting	0.662	0.731	0.475	0.071	0.571	0.816	0.814	0.725	0.512
	HI thumb-index	0.662	-0.100	0.200	0.038	0.511	1.000	1.000		-0.118
	HI middle-index		-0.754	0.741		0.200		0.543	-0.691	0.290
	ABSHI thumb-index	0.662	0.100	0.200	0.038	0.511	1.000	1.000		0.118
	ABSHI middle-index		0.754	0.741		0.200		0.543	0.691	0.290
Multitask	HI-multitask 1	0.827	0.221	0.635	0.337	0.612	0.896	0.709	-0.816	-0.367
	HI-multitask 2	0.827	0.231	0.617	0.369	0.565	0.927	0.732	-0.785	-0.193
	Consistency preference	Yes	No	Yes	No	Yes	Yes	Yes	Yes	No

The main grip morphologies used were thumb+index (45.15%) and index+middle (38.74%). 14.31% used other grips and 1.80% of the grips were not visible to observers. There is a significant difference between grip morphology used (thumb+index, index+middle and other grip morphologies) and hand preference (left or right) (χ^2=17.463; p=0.000). Analyses of adjusted residuals showed that thumb+index elicit right-hand use (AR = 2.7), while middle+index elicit left-hand use (AR = 4.2). There were no differences between thumb-index and middle-index grip morphologies in direction (HI thumb-index = 0.222, HI middle-index = 0.210; $Z = -0.700$, $p = 0.484$) or strength (ABSHI thumb-index = 0.378, ABSHI middle-index = 0.517; $Z = -0.840$, $p = 0.401$) of hand preference. There is no significant

					Average values			One sample *t*-test			Lateralized
Tico	Toni	Toto	Víctor	Waty	Mean value	SE	SD	*t*	df	*p*	L/R/A
0.000	0.000	0.000	0.000	0.000							
L	R	R	R	L							4/10/0
-0.304	0.806	1.000	0.410	-1.000	0.352	0.211	0.789	1.670	13	0.119	
-0.139	0.908	1.000	0.560	-1.000	0.416	0.198	0.470	2.101	13	0.056	
0.304	0.806	1.000	0.410	1.000	0.811	0.058	0.218				
34.80	90.30	100.00	70.49	0.00	67.61	10.544	39.452				
227	268	123	122	438	194.93	32.600	121.978				
-0.333	1.000	1.000	0.366	-1.000	0.388	0.200	0.749	1.939	13	0.075	
0.091	1.000	1.000	1.000	-1.000	0.510	0.194	0.724	2.636	13	0.021	
0.280	0.634	1.000	0.579	-1.000	0.453	0.203	0.759	2.234	13	0.044	
-0.594	1.000	1.000	0.296	-1.000	0.312	0.221	0.825	1.413	13	0.181	
-0.304	1.000	1.000	0.400	-1.000	0.333	0.228	0.822	1.459	12	0.170	
	-0.156	1.000		-1.000	0.468	0.292	0.773	1.602	6	0.160	
0.304	1.000	1.000	0.400	1.000	0.827	0.065	0.235				
	0.156	1.000		1.000	0.798	0.125	0.332				
2	3	2	2	4	2.14	0.254	0.949				
0.000	0.000	0.000	0.029	0.426							
R	L	R	R	A							3/9/2
0.855	-0.480	0.500	0.214	-0.089	0.221	0.133	0.499	1.654	13	0.122	
0.500	0.467	0.214	0.855	0.653	0.464	0.071	0.264				
92.73	26.00	75.00	60.71	45.54	60.96	6.673	24.970				
110	100	100	112	101	119.64	5.148	19.262				
0.828	-0.097	0.233	0.333	0.298	0.305	0.089	0.333	3.429	13	0.004	
0.885	-0.652	0.900	0.143	-0.591	0.215	0.172	0.644	1.248	13	0.234	
0.828	0.097	0.233	0.333	0.298	0.341	0.078	0.292				
0.885	0.652	0.900	0.143	0.591	0.611	0.067	0.250				
	-0.615	0.120	0.068	-0.103	0.222	0.140	0.485	1.587	11	0.141	
0.841	-0.091	0.582	0.440		0.210	0.177	0.560	1.187	9	0.266	
	0.615	0.120	0.068	0.103	0.378	0.105	0.364				
0.841	0.091	0.582	0.440		0.517	0.081	0.255				
0.275	0.163	0.750	0.312	-0.545	0.286	0.141	0.529	2.025	13	0.064	
0.358	0.214	0.750	0.387	-0.545	0.318	0.136	0.511	2.332	13	0.036	
No	No	Yes	Yes	No							

group level right-handedness for thumb-index grip ($t(11) = 1.587$; $p = 0.141$), nor for middle-index grip ($t(9) = 1.187$; $p = 0.266$) morphologies.

The main postures adopted were sitting (58.62%) and quadrupedal (40.66%); other postures adopted: 0.72% (prostrate, bipedal). There is a relationship between posture (quadrupedal, sitting, and others) and hand preference (right, left; $\chi^2 = 15.578$; $p = 0.000$). When analyzing adjusted residuals, quadrupedal posture (AR = 3.9) elicited right-hand use, while sitting (AR = 3.9) elicited left-hand use. There was no difference between quadrupedal and sitting postures in the direction (HI quadrupedal = 0.305, HI sitting = 0.215; $Z = -0.596$, $p = 0.551$); however, there are differences in the strength (ABSHI quadrupedal = 0.341, ABSHI sitting = 0.611;

Table III Spearman correlations between HI values across test sessions

	HI Test 1	HI Test 2	HI Test 3	HI Test 4
HI Test 1		0.0836**	0.696**	0.922**
HI Test 2			0.594*	0.757**
HI Test 3				0.773**

Significant at **$p<0.01$; *$p<0.05$.

$Z = -2.524$, $p = 0.011$) between both postures. There is significant group level right-handedness for quadrupedal posture ($t(13) = 3.429$; $p = 0.004$), but not for sitting posture ($t(13) = 1.248$; $p = 0.234$).

Multitask Assessment

There is no significant difference in the direction of manual preference (handedness index) between both tasks ($Z=-0.973$; $p=0.331$). Nevertheless, there are significant differences in the strength of hand preference (ABSHI): TUBE shows higher than REACH ($Z=-2.605$; $p=0.009$; Fig. 1). There is no significant correlation between individuals HI for REACH and TUBE ($R_s = 0.372$; $p = 0.190$), nor between absolute HI values in both experiments ($R_s = 0.077$; $p = 0.793$). Overall, 57.14% of the sample is consistent and 42.86% is not consistent in their manual preferences.

To evaluate the existence of handedness at population level we performed different analyses. First, we calculated an individual mean HI for both tasks (multitask HI) based on SUMHI TUBE and HI REACH; we called this new index HI multitask 1 (HI-MT1). Second, we calculated an individual mean HI for both tasks based on MHI TUBE and HI-REACH; we called this new index HI multitask 2 (HI-MT2). We performed a 1-sample *t*-test for MT1 and MT2. The sample was right-handed at the population level on MT2 ($t = 2.332$; $p = 0.036$), and there was borderline significance for right-hand preference at the population level for MT1 ($t = 2.025$; $p = 0.064$).

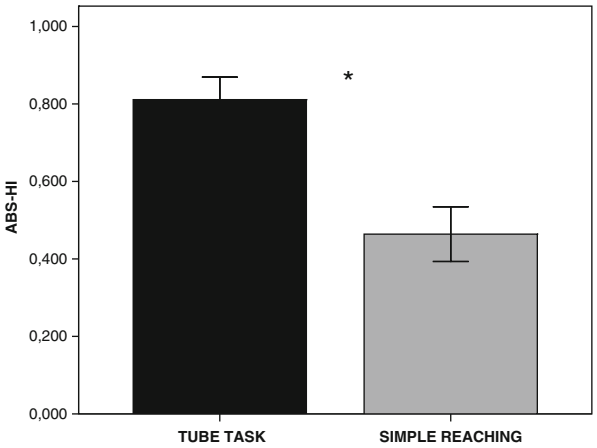


Fig. 1 Mean (±SE) of hand preference strength for tube task and simple reaching task. Significance of Wilcoxon test: *$p<0.05$.

Table IV Distribution of significant manual preferences of subjects across tasks

	Always left	Significant left	Ambipreferent	Significant right	Always right	*n* total
Tube task	1	3	0	8	2	14
Simple reaching	0	3	2	9	0	14

Discussion

Results for the tube tasks indicate that the focal sample showed clear and strong hand preferences at the individual level for this task (Table IV). Most of the individuals are right-handed, but we did not detect asymmetry at the population level, except for 2 tests (test 2 and test 3), where they showed a right-handedness at the group level. Although failing to demonstrate any statistical significance to conclude that there was group asymmetry for the task, we consider it important that 100% of the sample were lateralized. We support the suggestion of Hopkins that chimpanzees could be preferentially right-handed for a tube coordinated bimanual task (Hopkins 1995; Hopkins *et al.* 2004). Similarly as in previous studies (Hopkins and Cantalupo 2003; Hopkins *et al.* 2001), consistency in hand use was consistent throughout sessions, showing a high correlation within the 4 tests. The strength of hand preference is high, and it may indicate that the tube task elicits a strong lateralization in individuals, irrespective of the directionality, as some other authors have argued (Hopkins *et al.* 2003a). The need to coordinate both hands in a task produces an increase in the number of individuals lateralized and in the strength of lateralization of chimpanzees of our sample. Researchers have observed the effect in other species such as *Cercocebus torquatus torquatus* (Blois-Heulin *et al.* 2006), *Cebus apella* (Lilak and Phillips 2008; Spinozzi *et al.* 1998; Spinozzi and Truppa 1999; Westergaard and Suomi 1996), *Cercopithecus neglectus* (Schweitzer *et al.* 2007; Trouillard and Blois-Heulin 2005), *Cercopithecus aethiops* (Harrison and Byrne 2000), besides *Pan troglodytes* (Colell *et al*. 1995b).

We detected no differences between digital or tool techniques and hand preference, but finger 2 elicits right-hand use, the other fingers elicit left-hand use, and tools seem to have no effect on direction of preference, possibly because both left- and right-handed individuals use the technique. The preference in index finger matches that of other studies in chimpanzees (Hopkins 1995) and other primate species (Schweitzer *et al.* 2007; Spinozzi *et al.* 2007), but this is the first time that tool extraction is assessed for the tube task in chimpanzees. Distal movements in fingers and hands in bimanual actions could have an effect on the expression of manual laterality. As some authors have argued (Brinkman and Kuypers 1973), distal movements require a more frequent use of the contralateral hemisphere, which may explain our results. One of the main differences between our study and those of others with the tube task is the use of tools. Taking into account that tool use requires additional complexity, our results show a high handedness index value and absolute HI value.

For simple reaching, most of the subjects showed significant hand preferences. This result is similar to others obtained for the same task (Colell *et al*. 1995a). Some

authors argued that this is a low-level task (Fagot and Vauclair 1991), and we could expect weak preferences and a symmetrical distribution between right- and left-handed individuals. Nevertheless, we found only 2 nonpreferent subjects, while the others were lateralized, but not at a group level preference for this task. Our results showed that thumb+index were the main grip morphology, as supported by other authors (Hopkins *et al.* 2002; Hopkins *et al.* 2005b), and reinforce the view that the results are not restricted to specific groups of individuals but instead appear to be consistent throughout colonies. Finally, as other authors have argued, postural effect influences hand preference (Colell *et al.* 1995a; Hopkins 1993). We found that quadrupedal and prostrate+bipedal posture elicit right-hand use, whereas sitting favors left-hand use, and for quadrupedal posture the sample was right-handed at the group level.

We found no preferences at the population level for either task. Nevertheless, if we compare hand preferences for TUBE and REACH, bimanual task elicited significantly greater individual asymmetries than the simple reaching task, a result that is beginning to be reported not only in chimpanzees but also in other nonhuman primate species (Colell *et al.* 1995b; Schweitzer *et al.* 2007; Spinozzi and Truppa 1999). Strong and overall lateralization in TUBE may be influenced by tool techniques and by the dominance of the index finger in food extraction. Also, the important lateralization found in REACH could be influenced when actions are performed in natural areas and environments that demand precision grips to reach food.

However, it is interesting that, despite the small sample size, comparison of the results for both tasks has revealed, for the first time in a naturalistic environment, that chimpanzees are right-handed at the population level (HI multitask 2) or at least they are borderline lateralized (HI multitask 1) for the right hand. The chimpanzees in our sample show handedness when the data are summed across multiple measures. In the words of McGrew and Marchant (1997), this would be considered evidence of true handedness (a human-like pattern) rather than task specific handedness (more of the nonhuman primate variety). However, we did not detect handedness at the population level in a earlier study of the same sample (Mosquera *et al*. 2007), which may be accounted for by methodological and procedure factors (Hopkins and Cantalupo 2005). Negative correlation and inconsistency between both tasks may suggest that manual laterality is a multidimensional trait, as some authors supported (Hopkins 2006; Wesley *et al.* 2002). According to their opinion, motor and neurological demands and requirements were different for the tube task and simple reaching (Hopkins and Pearson 2000). Even though simple tasks seem to provide little information about hand preferences (Blois-Heulin *et al*. 2006), we think that including variables such as grip morphology may be important in clarifying basic aspects of hand preferences. Therefore, although it seems that coordinating both hands increases the number of individuals lateralized and the strength of this lateralization (Fagot and Vauclair 1991), manipulative potential of primate hands and their gripping ability need to be taken into account. Both factors, bimanual coordination and fine grip, may have conditioned later tool use and tool making, their increasing complexity, and the greater lateralization of the hominid family. We also think that in spite of using experimental tasks, small samples are less sensitive than very large samples (Hopkins *et al.* 2005b; Hopkins *et al.* 2004) to

detect group bias, although our results seem to support those of Hopkins and colleagues.

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